



UCD Michael Smurfit
Graduate Business School

Machine Learning for Customers' Segmentation and Fraud Detection in the Insurance Sector

A Capstone submitted to University College Dublin in part fulfilment of the
requirements of the degree of M.Sc. in Business Analytics

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Dedication

We dedicate this work to our sponsor, Dr. Angel A. Juan, supervisor, Dr. Peter Keenan, and the team in Spain supervised by Dr. Angel A. Juan for their continued guidance and support during this project.

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List of Algorithms

ExtraTrees Classifier (ET)

Light Gradient Boosting Machine (LightGBM)

Categorical Boosting (CatBoost)

Extreme Gradient Boosting (XGBoost)

Preface

This report has been submitted to partially fulfil University College Dublin's MSc in Business Analytics requirements. My capstone project culminates in it since it employs machine learning for better customer selection choices during the insurance industry's Summer 2025 trimester.

The real business problem of an insurance company started this work: the need for portfolio profit maximisation with cost minimisation for incorrect classifications in customer acceptance. This project was an opportunity to integrate technical skills as we preprocessed data, selected features, developed models, plus evaluated them with a strong business perspective, ensuring solutions aligned with operational constraints and financial objectives.

During research academic supervisors and industry partners where applicable were in close collaboration with each other. They guided them to ensure that the methodology remained strict and that the evaluation criteria remained relevant. Because of them, people apply the findings in practical decision-making scenarios.

This project challenged as well as rewarded, so it was a valuable experience to balance statistical performance with overall business impact. This skill does remain critical for just about any data-driven role within the financial services sector. I am grateful I can apply the knowledge from the MSc programme to a real-world problem with concrete business implications.

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Furthermore, we extend our heartfelt appreciation to Dr. Angel A. Juan for providing the essential data that served as the foundation of this research. The access to real-world data has enriched the scope and relevance of this project, and we are grateful for your collaboration and support.

Last but not least, we are grateful to UCD Michael Smurfit Graduate Business School for presenting us with this opportunity to work on this capstone project. In closing, this project would not have been possible without the collective efforts and contributions of all those mentioned above. Thank you for being an integral part of our academic and personal growth.

Executive summary

This capstone project tackles a key challenge for an insurance company: accurately identifying profitable customers while minimising the rejection of those who would contribute positively to the portfolio. The company's existing rule-based system frequently accepts unprofitable customers, leading to substantial financial losses. The aim was to design, implement, and evaluate machine learning models that improve decision accuracy, boost portfolio benefit, and comply with the operational requirement of limiting customer exclusions to 2–5%.

The study used 15,486 historical customer records containing policy details and Key Performance Indicator (KPI) values. The workflow involved data preprocessing, splitting into train/validation/test sets, handling severe class imbalance via balanced class weights (effective `scale_pos_weight` = 4.17 for gradient boosting models), and selecting the nine most important predictors using ExtraTrees feature importance analysis. This produced a reduced, deployment-ready dataset, optimised for efficiency and predictive power.

Four models were developed and compared: ExtraTrees (baseline), LightGBM, CatBoost, and XGBoost. Probability thresholds were tuned on the validation set to meet the exclusion target while maximising portfolio KPI benefit. Models were assessed using both statistical metrics, ROC AUC, confusion matrix, classification report, and business metrics, such as exclusion percentage, portfolio benefit after model, benefit per customer, and error-cost analysis.

LightGBM emerged as the best-performing model, delivering a ROC AUC of 0.71 on the test set, excluding 4% of customers, and increasing estimated portfolio benefit from €406,006 to €1,044,221, a 157% uplift. Average benefit per customer improved from €26.22 to €67.43. Error-cost analysis highlighted an asymmetry in financial risk: while erroneously excluded profitable customers caused an average loss of €248 each, erroneously accepted unprofitable customers cost around €1,000 on average, with extreme cases exceeding, €175,000.

These results demonstrate that a business-aware machine learning model, particularly LightGBM, can significantly enhance underwriting decisions. Its deployment, alongside continuous monitoring and possible secondary review processes, could materially improve portfolio profitability while managing operational risk.

List of important abbreviations

AI	Artificial Intelligence
AUC	Area Under the Curve
CI/CD	Continuous Integration / Continuous Deployment
ET	ExtraTrees Classifier
GBM	Gradient Boosting Machine
KPI	Key Performance Indicator
LightGBM	Light Gradient Boosting Machine
CatBoost	Categorical Boosting Algorithm
XGBoost	Extreme Gradient Boosting Algorithm
ML	Machine Learning
ROC	Receiver Operating Characteristic
SMOTE	Synthetic Minority Over-sampling Technique
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative

Chapter 1

1.1 Introduction

Spain's insurance market sits in an interesting, sweet spot: it is large enough to move national macro-numbers yet still small enough that one or two percentage points of performance show up on each carrier's bottom line. Latest figures from (Manzano.D, 2024) put 2023 premium income at €76 billion, a new record and more than 18 % compared to the previous year. roughly making up 5 % of total Spanish GDP. The non-life sector including motor, property, health makes up just over half of that total, with motor lines alone accounting for more than a third of written premiums.

That growth comes with headwinds. Inflation has pushed up claims-repair costs; regulators keep a close eye on consumer pricing; and competition from bancassurance and direct-to mobile challengers squeezes operating margins. To respond, Spanish insurers are putting real money into advanced analytics.

Fraud is a stubborn, often silent drag on profit. The 2022 global FRISS report found that claims adjusters now suspect fraudulent elements in one out of every five P&C claims (20 %), up from 18 % in 2020 (Friss, 2022). Although the study is global, interviews with our industry sponsor confirm the Spanish ratios are “in the same ball-park.” At a motor portfolio loss ratio of 70 %, even a single hidden percentage-point of fraud wipes out several million euros of annual profit and because fraud rings learn fast, manual rules rarely keep pace.

Regulation adds another layer of urgency. The newly finalized EU AI Act places fraud detection engines squarely in its ‘high-risk’ category, meaning any model we build must expose clear explanations, undergo bias testing and keep full audit trails (Commission, 2024). Those obligations go together with Spain's existing Solvency II and GDPR regimes. Taken together, they reward models that are not only accurate but transparent and low latency. Precisely the space where gradient-boosted trees shine.

Finally, the data landscape is evolving rapidly. Spain's Directorate-General for Traffic publishes yearly open datasets on accident density and vehicle

registrations, while the National Statistics Institute releases granular socio-economic indices down to the municipal level (Observatory, 2022). Insurers have begun to enrich their policy files with these public feeds, opening the door to more nuanced risk scores than traditional age-and-postcode pricing.

We are looking at a capital-heavy, regulation-tight market that desperately needs higher resolution views of customer risk and sharper fraud filters. The pieces of rich internal data, external open datasets, and maturing Machine learning toolkits are finally in place to make that happen.

1.2. Justification

Three factors make this problem economically material. First, legacy rules leave substantial value uncaptured: on our sponsor's extract, the realized benefit prior to modelling is €0.91 million versus a theoretical €12.24 million if all profitable customers were accepted and all loss-makers were filtered, implying a large recoverable upside. Second, misclassification is costly on the tail: a small number of severely loss-making policies can erase thin margins in motor portfolios, a concern echoed in fraud-related literature where adjusters report a material share of suspicious claims. Third, compliance and trust requirements in Spain and the EU favor transparent, auditable models with clear rationales for decisions at quote time; tree-based gradient boosting with post-hoc explanations.

1.3 Objective

Our team's business objective is to reduce the cost the insurer loses to undetected fraud while, at the very same time, giving each prospective customer a risk score that supports fair-but competitive pricing, while providing the real time results that online quoting demands. We target higher recall for the loss-making class at controlled false-positive rates aligning with a 4–6% exclusion band agreed with the sponsor, and we measure success in euros via a benefit-recapture KPI: the model's estimated total benefit on the test population versus both the realized status quo and the theoretical ceiling. Industry surveys show that eight in ten Spanish insurers now experiment with Artificial Intelligence, yet suspected fraud still hovers at about 20 % of all motor-claim submissions; most carriers capture barely half of those cases in real life. Missing the other half is expensive: finance

managers at our sponsor estimate that every single percentage-point gain in fraud recall is worth roughly €0.7 million a year in prevented pay-outs.

To deliver that value we have set three concrete targets. First, raise fraud-class recall from the current 58 % ExtraTrees baseline to at least 75 % while holding precision above 65 %. Secondly to return a segmentation-driven price adjustment within a shorter period so the quotes are produced in real-time. Achieving those numbers will rely on the pipeline we have already scoped: SMOTE-balanced training folds, gradient-boosted models including LightGBM, and XGBoost tuned for recall, and a machine learning operations layer that tracks drift from the required output and retrains quarterly thus promotes the new version only when it is demonstrably better.

1.3. Business problem

Our challenge in Spain's auto-insurance sector is to predict, using only minimal application data available at quote time and without any prior behavioral history at the carrier, whether a new policyholder will be profitable (class 0) or loss-making (class 1), so underwriting can be more precise and margins protected.

1.4 Scope

Data and features. We work with the sponsor's cleaned motor-policy file (51,618 rows; 196 columns after dropping unavailable closed-book aggregates). SINCO variables are central, exposure time, claim frequencies and severities, vehicle counts by type, and consultation flags, supplemented by vehicle value, driver tenure, channel codes, and risk flags. We audit types, ranges, duplicates, and confirm zero missingness post-pruning. For model parsimony and regulatory clarity, we later restrict to a compact, high-signal subset informed by feature importance and statistical screening

Labels and imbalance. The target is constructed from BMA_corregido: values > 0 map to class 0 (accept), ≤ 0 to class 1 (exclude). Imbalance is managed through stratified splits and sampling on the training fold using class-weighted learners.

Model family. We benchmark strong tabular learners: Extra Trees, baseline and early-stopped, XGBoost, LightGBM, and Cat Boost. Gradient boosting

frameworks are state of the art for structured data, with proven scalability and regularization controls; they are also compatible with feature-attribution tooling for post-hoc explanations.

Early stopping and thresholding. To reduce overfitting and converge on the most predictive ensemble size, we employ validation-based early stopping where supported, then re-fit at the selected iteration on the training split. Because business constraints limit the acceptable exclusion rate, we calibrate the decision threshold on the validation set to hit a 4–6% reject band and then hold that threshold fixed on the test set; we report full ROC-AUC and “partial-AUC” in the low-FPR region that matches operational tolerances.

Evaluation and economics. Beyond ROC-AUC and class-wise precision/recall/F1, we compute three benefit figures on each split:

- i. theoretical maximum (sum of positive KPI),
- ii. realized pre-model benefit, and
- iii. model benefit given the selected customers.

The difference between the realized pre-model benefit and the model benefit given the selected customers is the benefit recaptured by the model; we report both total euros and per-customer lift. Error analysis highlights false negatives which consist of the profitable customers wrongly excluded via lost-benefit histograms and quantiles, and false positives which consist of loss-makers wrongly accepted via cost tails.

Governance and compliance. We design for explainability and auditability: feature-importance screens guide feature reduction; per-prediction attributions are available for review; and drift checks including PSI on top variables compare train vs. validation/test to reduce the risk of data-shift surprises at deployment. The regulatory backdrop, EU AI Act obligations for high-risk insurance systems and GDPR, motivates these controls.

Chapter 2 - Literature Review

The insurance sector has adopted supervised learning to support underwriting, pricing, and fraud screening where the target is binary and the feature space is tabular and mixed-type. Gradient-boosted decision trees and ensembles remain the reference class of models for tabular risk prediction thanks to their fit to non-linear interactions, tolerance to monotone transformations, and speed at inference time. XGBoost introduced a highly efficient tree-boosting system that scales well on large matrices and offers regularisation to curb overfitting (Chen, 2016). LightGBM refined this approach with histogram-based splits and leaf-wise growth that reduce training time without a material loss in accuracy on many tabular tasks (Ke, 2017). CatBoost added native handling of categorical variables via ordered target statistics to reduce target leakage and overfitting in high-cardinality settings (Guolin Ke, 2017). Extremely Randomized Trees (ExtraTrees) offer a contrasting ensemble that injects randomness at split selection to lower variance and provide useful feature-importance diagnostics, often at modest computational cost. These families of models are widely reported in insurance risk scoring and fraud detection case studies and reviews, where tabular structured data dominate practical deployment (Ahmed, 2016).

A central challenge in underwriting datasets is class imbalance. Loss-making policies or confirmed frauds form a minority of cases, yet missing them carries significant financial cost. Cost-sensitive learning complements resampling by weighting training loss with class- or case-specific costs so that the learned decision surface reflects business penalties rather than sample frequencies (Elkan, 2001). Insurers frequently combine these ideas: light oversampling to stabilize training, class-weighted losses to reflect claim costs, and threshold tuning at evaluation to match operational constraints.

Evaluation under imbalance requires care. ROC-AUC can mask poor minority performance when prevalence is low; precision–recall curves and PR-AUC are often more informative for rare-event detection (Saito, 2015). When business rules cap the allowable false-positive rate, partial AUC over a restricted FPR range provides a clearer signal than full AUC (Facwett, 2006). Probability calibration also matters if outputs will be thresholded by expected profit; isotonic

regression and Platt scaling are standard post-hoc methods to convert scores into better calibrated probabilities (Zadrozny, 2002). Combining calibrated probabilities with cost matrices supports threshold selection that maximizes expected benefit rather than a purely statistical metric (Elkan, 2001).

Feature selection and model transparency are important in regulated contexts. Global feature importance from tree ensembles offers a first screen for dimensionality reduction and sanity checks about signal sources (Pierre Guerts, 2006). For case-level explanations, Shapley additive values provide consistent, local attributions that sum to the prediction, which helps audit high-stakes automated decisions and communicate drivers to underwriting teams (Lundberg, 2017). These tools are now widely recommended in credit and insurance deployments where traceability is required.

Monitoring for drift across data splits and over time is a separate thread in the literature. The Population Stability Index (PSI) is widely used in financial model risk management to compare score or feature distributions between a reference sample and a new sample. Rules of thumb interpret $PSI < 0.1$ as stable, $0.1-0.25$ as a warning band, and >0.25 as a shift that warrants investigation and potential recalibration (Josh Brownlow, 2015). PSI is simple to compute on numeric binning's and supports routine governance dashboards after deployment.

Insurance-specific data and context shape feature engineering. Spanish motor data often incorporate SINCO fields, a national information system coordinated within the sector that records motor claim and policy history for applicants. These variables capture exposure time, claim frequency and severity indicators, and vehicle counts that are linked to risk and loss experience. Public documentation describes SINCO as the shared “siniestralidad” information layer used by carriers to verify history at underwriting. Such fields align with long-standing actuarial practice that uses rating variables with demonstrated correlation to expected claim costs (Institute, 2019).

Recent policy and regulation impose requirements on automated decision systems that are pertinent for motor underwriting. The EU AI Act classifies many insurances risk-assessment systems as high-risk and sets expectations for risk management, data governance, transparency, and human oversight (EUR-LEX,

2024). GDPR Article 22 and related guidance limit solely automated decisions that have legal or similarly significant effects and motivate the need for explanation, contestability, and bias testing (Board, 2018). These frameworks have pushed practitioners toward models that are not only accurate but also explainable, auditable, and fast enough for near real-time quotes.

A final stream of work focuses on profit-based evaluation rather than error rates. Studies in credit and insurance propose objective functions and metrics that directly optimise or report expected monetary return under user-provided costs and benefits, which often generate different thresholds and rankings than accuracy-based tuning. This orientation is pertinent when the goal is to filter a small share of applicants to avoid large expected losses while retaining most profitable policies. It supports dashboards that translate ROC and PR trade-offs into euros per policy and total benefit captured.

The methods above map directly to the choices in our pipeline. We work in a tabular setting with many engineered variables, so gradient-boosted trees and ExtraTrees are natural baselines and finalists (Chen, 2016). Imbalance is non-trivial in our data, which motivates class weighting, and careful threshold selection by expected benefit (Elkan, 2001). We report full ROC-AUC together with PR-AUC and partial AUC in a low-FPR band that reflects underwriting constraints (Saito, 2015). We use global feature importance to cut the feature space to explain individual predictions to non-technical stakeholders (Lundberg, 2017). Split-stability checks with PSI help confirm that train, validation, and test folds are comparable, which reduces the risk that our validation gains are driven by sample quirks rather than signal. Lastly, we present benefit-oriented summaries to tie model performance to business impact where the decision is exclusion at a small rate versus profit preservation on many applicants (Franco Garrido, 2018).

Chapter 3 – Methodology

In the highly competitive insurance sector, accurately distinguishing “high-ROI” customers from those likely to incur losses is essential for pricing, fraud prevention, and resource allocation. Our approach tackles two core challenges: severe class imbalance—where profitable policies vastly outnumber loss-making ones and the need for both interpretability and real-time performance. By combining resampling, dimensionality reduction, unsupervised profiling, and a diverse ensemble of classifiers, we build a pipeline that not only maximizes predictive power $\text{ROC-AUC} > 0.82$ on validation but also delivers clear business insights via key-performance metrics (KPI) and cluster analysis.

3.1 Overall Pipeline Architecture

Our methodology follows a four-stage, business-driven pipeline designed to ensure interpretability, operational feasibility, and direct alignment with profitability objectives. The process begins with data preprocessing and leakage control, where we clean the raw policy dataset, standardize column names to an alphanumeric format, and remove aggregate fields such as `PrimaTotalPoliza`, `ComisionTotalPoliza`, `SiniestralidadTotalPoliza`, and `ExposicionTotalPoliza`, as these are not available at the time of new customer evaluation and could cause data leakage. We also exclude records without SINCO information, which reduces the dataset from 116,934 to 51,618 policies. The target variable (`true_class`) is derived directly from the KPI, where a value of **0** represents a profitable customer ($\text{KPI} > 0$) and a value of **1** represents a loss-making customer ($\text{KPI} \leq 0$). After filtering, no missing values remain, eliminating the need for imputation.

The second stage addresses the class imbalance problem, as the dataset is heavily skewed towards profitable customers. Instead of applying synthetic oversampling methods such as SMOTE, we handle imbalance through computed class weights, which are passed directly to the models. For `ExtraTrees`, we use balanced class weights, and for gradient boosting algorithms such as `LightGBM` and `XGBoost`, we set the `scale_pos_weight` parameter to approximately 4.17. This ensures that the minority “exclude” class is adequately represented during training without introducing artificial data points or distorting the original distribution.

In the third stage, we perform feature selection and supervised modeling. We first train an ExtraTrees classifier on all features (excluding KPI and the target) to obtain feature importance rankings. From this ranking, we select the top nine predictors—id55, Cliente_Diverso, AnyoPoliza, TiempoMedioSINCO, Antigüedad_Carnet, Edad, AntigüedadVehículo, ValorVehículoMercado, and SINCOMatriculaIncluida. This reduced, deployment-ready feature set helps minimize noise, improve generalization, and ensure consistent input requirements for production deployment. Using this optimized feature subset, we train four supervised learning models: ExtraTreesClassifier (used both as a baseline and as the feature importance source), LightGBM, CatBoost, and XGBoost. These models are trained with appropriate class balancing parameters to address the skewed distribution in the data.

The final stage focuses on business-aware threshold tuning and benefit analysis. Rather than relying on the default probability threshold of 0.5, we tune the classification threshold on the validation set with two objectives: first, to keep the exclusion rate between 2% and 5% in order to meet customer retention targets; and second, to maximize the total portfolio benefit, defined as the sum of KPI values for accepted customers. This ensures that the final model is optimized not only for statistical performance but also for real-world business value. Once the optimal threshold is determined, we evaluate the model on the test set using both standard metrics (ROC AUC, confusion matrix, classification report) and business metrics (percentage of customers excluded, portfolio benefit after applying the model, benefit per customer, and detailed error-cost analysis comparing lost benefit from erroneously excluded customers versus erroneously accepted ones). Predictions and associated KPIs for test customers are then exported to CSV for downstream use, and the best-performing model is serialized with joblib for deployment. The final pipeline is fully documented with its required feature set, ensuring smooth integration into the underwriting workflow and consistent alignment with profitability goals.

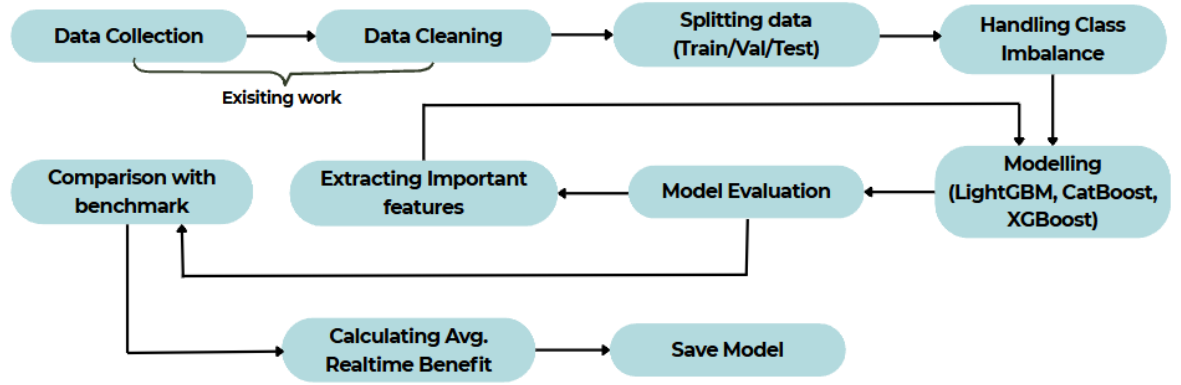


Figure 1: Process Diagram

3.2 Data Collection and Cleaning

The dataset was provided by our industry partner and consisted of anonymized customer records, enriched with historical policy, claims, and vehicle-related information. This initial collection formed the raw input for our pipeline.

Data cleaning involved handling missing values, correcting inconsistent data types, and removing duplicate or corrupted records. Categorical features were encoded, numeric features were standardized where appropriate, and outliers were examined to ensure they did not result from data entry errors.

3.3 Exploratory data analysis

3.3.1 Quick audit: structure, types, memory, completeness

We began by auditing the raw modelling table to verify basic integrity and reproducibility. After pruning policy-level aggregates not available at quote time, removing records without SINCO history, and standardising column names, the analytical dataset contained 51,618 rows $\times$ 196 predictors plus the binary target `true_class`. Pandas reported dtypes: 196 float64 and 1 int32 (`true_class`), which is advantageous because most features are already numeric and require minimal encoding. Memory footprint at this stage was $\sim$ 79.8 MB, small enough for fast iteration on a laptop while preserving full fidelity of the original extracts.

A completeness check confirmed no missing values across the retained columns (see Figure 2). This eliminated the need for imputation and ensures that downstream

comparisons of models are not confounded by imputation choices. We also verified that duplicate policy rows were absent after de-duplication by policy ID, so each row represents a unique policyholder-period observation.

```
Columns with more missing data:
id17          0
TotalMatriculasSINCO  0
PerteneceSINCO  0
TiempoMedioSINCO  0
TiempoSeguroSINCO  0
dtype: int64

Total number of missing data: 0
```

Figure 2: Missing values observed in dataset

3.3.2 Target prevalence and KPI distribution

Figure 3 shows the class balance of `true_class` (0 = profitable/accept, 1 = loss-making/exclude). The portfolio is imbalanced with 80.7 % class 0 and 19.3 % class 1. This imbalance explains why naïve accuracy can be misleading and motivates the use of ROC-AUC, cost-sensitive training and threshold tuning.

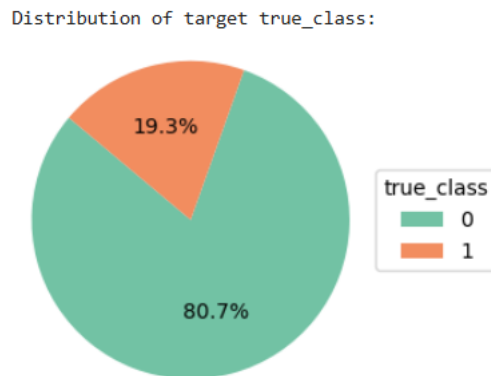


Figure 3: Class distribution pie (80.7 % vs 19.3 %).

Because profitability is the business objective, we examined the financial KPI `BMA_corregido`. The overall histogram and boxplot (Figures 2 and 3) reveal an extremely right-censored, left-skewed distribution with a long negative tail. Most observations lie close to zero or modestly positive; a minority of policies generate large negative outcomes. This heavy-tail behaviour is typical of insurance loss data and has two practical implications:

1. Robustness matters. Median and IQR are more informative than mean/std for central tendency.
2. Cost-aware modelling. False negatives (erroneously excluding profitable customers) have small euro impact; false positives (accepting truly loss-making policies) can be extremely costly. This asymmetry supports threshold tuning and sample-weighting by KPI magnitude in training.

Boxplot of KPI in dataset X:

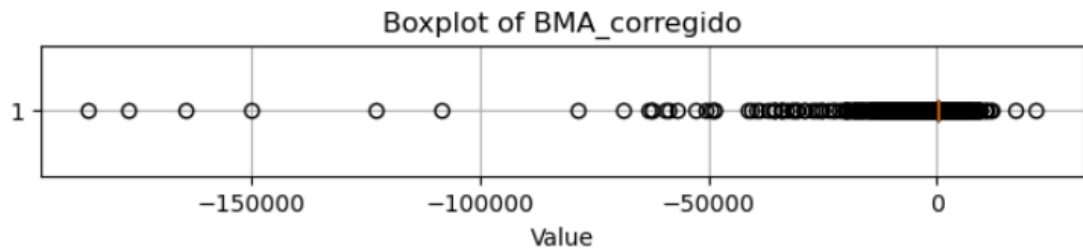


Figure 4: KPI by class (boxplots for true_class = 0 vs 1).

This boxplot stratified by true_class show that the excluded class (1) concentrates most large negative KPI outliers, validating our target definition from a financial perspective; the accepted class (0) centres around modestly positive KPI with occasional small negatives.

3.3.3 Feature–target screening

3.3.3.1 Point–biserial correlation

To quantify linear associations between individual numeric predictors and the binary target, we computed the point–biserial coefficient r_{pb} . This statistic is equivalent to the Pearson correlation between a numeric feature and a 0/1 target; it measures how strongly feature levels separate the two classes. Large absolute r_{pb} with small p-values indicates discriminative power. See *Figure 5: Point-Biserial ranking*

	feature	r_pb	p_value
157	Cliente_Diverso	-0.148844	1.934763e-160
3	id55	0.147162	7.656741e-157
126	TiempoMedioSINCO	-0.095597	6.936097e-67
136	AnyoPoliza	-0.082838	1.277723e-50
146	id72_2	0.065260	4.929647e-32
178	id54_SEMESTRAL_DOMICILIADO	0.061584	1.056380e-28
142	ContrataLunas	0.059444	7.483376e-27
122	Antiguedad_Carnet	-0.058342	6.338684e-26
182	Subramo_MOTOCICLETA	-0.055187	2.305768e-23
11	ValorVehiculoMercado	0.050894	4.185624e-20
10	id70	0.050552	7.420837e-20
164	id13_H	0.049366	5.235600e-19
140	id71	0.048894	1.125959e-18
125	PerteneceSINCO	-0.047195	1.665143e-17
12	SINCOTotalVehiculos	0.045624	1.854165e-16

Figure 5: Point Biserial ranking

Signs indicate direction, in the case of higher Cliente_Diverso values associate with class 0), not causality. Magnitudes near 0.05–0.15 are typical when predicting noisy behavioural outcomes; the small p-values reflect the large sample size rather than huge effects. This screen is helpful for ranking features but does not capture non-linearities or interactions.

3.3.3.2 Mutual information (non-linear dependence)

Table 1: Mutual Information scores(top 20)

Variables	Mutual Information
id55	0.028693
Cliente_Diverso	0.010377
SINCOTotalVehiculos	0.010286
INE_027	0.01028
AnyoPoliza	0.009852
ScoreCanal	0.007567
DGC_011	0.006834
DGC_020	0.00666
OSM_001	0.006319
INE_047	0.006201
id13_E	0.006019
INE_029	0.005926
id58_3	0.005752
INE_052	0.005718
TiempoMedioSINCO	0.005659
id54_SEMESTRAL_DOMICILIADO	0.005578
riesgo_3	0.005492
id70	0.005451
DGC_004	0.005183
FrecuenciaDanyo_Materiales	0.005078

To complement rpbr, we estimated mutual information (MI) between each feature and the binary target. MI captures any form of statistical dependence (not just linear). The leading variables by MI included:

Although absolute MI values are small (common in high-variance human/claim processes), the relative ranking broadly agrees with point–biserial results. This triangulation increased confidence in the cross-model feature importance consensus used later to reduce dimensionality to nine predictors without sacrificing accuracy.

3.3.4 Dependence between predictors and multicollinearity

We next examined inter-feature dependence to understand redundancy and potential instability for linear models.

3.3.4.1 Spearman correlation heatmap

Figure 6 plots a Spearman correlation heatmap for the top 20 point–biserial features. Several blocks are evident:

- A SINCO cluster (FrecuenciaSiniestroSINCO, FrecuenciaDanyo_MaterialesSINCO, NV_AutomovilSINCO, SINCOTotalVehiculos, TiempoSINCO_Anos, PerteneceSINCO) showing moderate to strong positive associations. These reflect the shared origin in the claims registry and overlapping operational definitions (more years in SINCO → more opportunities for recorded events).
- A pricing/contract cluster (id54_SEMESTRAL_DOMICILIADO, ContrataLunas, id71, id70) with moderate correlations, plausibly tied to payment modality and optional coverages.
- Weak correlations among Cliente_Diverso, AnyoPoliza, and age-related variables, suggesting complementary information.

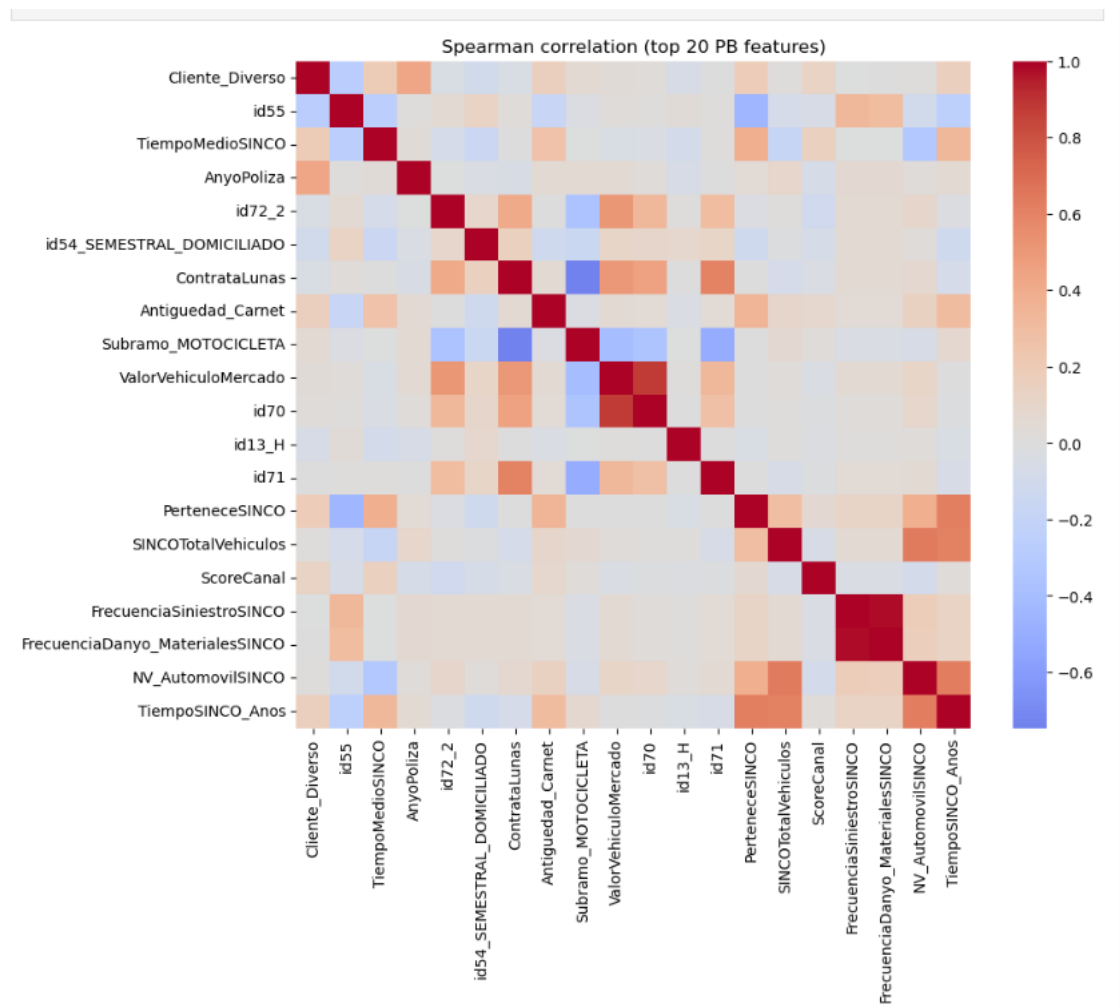


Figure 6: Spearman Correlation of top 20 features

3.3.4.2 Variance Inflation Factor (VIF)

To quantify collinearity for linear specification risk, we computed VIF on a subset of interpretable predictors. Two SINCO frequency variables showed $VIF = 20$ ($FrecuenciaSiniestroSINCO = 19.88$; $FrecuenciaDanyo_MaterialesSINCO = 19.54$), indicating severe multicollinearity if used together in a linear model. Several others were moderate including $ContrataLunas$ 4.10; $id71$ 3.55; $NV_AutomovilSINCO$ 3.36. VIF values near 1–2 are benign ($Cliente_Diverso$ 1.35; $id72_2$ 1.28).

Implication. For tree-based models (ExtraTrees, XGBoost, LightGBM), multicollinearity is less problematic; trees naturally select splits from redundant covariates. For linear or GLM baselines, the high VIFs motivate feature selection/regularisation and using only one of the highly collinear SINCO frequencies. This rationale underpinned our decision to reduce input dimensionality via cross-model importance union before final training.

	feature	VIF
0	const	758765.575103
17	FrecuenciaSiniestroSINCO	19.878026
18	FrecuenciaDanyo_MaterialesSINCO	19.538599
7	ContrataLunas	4.099118
13	id71	3.554880
19	NV_AutomovilSINCO	3.362434
20	TiempoSINCO_Anos	2.959706
9	Subramo_MOTOCICLETA	2.460416
15	SINCOTotalVehiculos	2.235240
11	id70	2.186580
3	TiempoMedioSINCO	2.130229
14	PerteneceSINCO	1.798698
10	ValorVehiculoMercado	1.602948
1	Cliente_Diverso	1.354646
5	id72_2	1.279807

Figure 7: VIF by feature (subset).

3.3.5 Data-set stability and leakage checks

Robust generalisation requires that training, validation and test splits are drawn from the same population and that no target leakage is present. To assess split stability, we computed the Population Stability Index (PSI) for the top features comparing train to validation and train to test distributions. All PSI values were well below 0.1 (most 0.000–0.008; see *Figure 8*). Conventional rules of thumb classify $\text{PSI} < 0.1$ as no shift, $0.1\text{--}0.25$ as moderate shift, and > 0.25 as material shift. Our results therefore indicate stable sampling and no detectable drift across splits.

This finding is consistent with our governance constraint: we removed all features unavailable at quote time including closed-book aggregates, normalised column names, and verified that time-dependent fields are included only as contemporaneous attributes rather than future outcomes. Together, these checks provide evidence against label leakage.

	feature	PSI(train→val)	PSI(train→test)
0	Cliente_Diverso	0.001	0.000
1	id55	0.001	0.001
2	TiempoMedioSINCO	0.004	0.001
3	AnyoPoliza	0.004	0.000
4	id72_2	0.000	0.000
5	id54_SEMESTRAL_DOMICILIADO	0.000	0.000
6	ContrataLunas	0.000	0.000
7	Antiguedad_Carnet	0.008	0.001
8	Subramo_MOTOCICLETA	0.000	0.000
9	ValorVehiculoMercado	0.007	0.002

Figure 8: Population Stability Index for top features (train→val, train→test).

3.3.6 Domain-specific EDA: SINCO features

SINCO variables (Spanish claims registry) were central to discrimination. In the point–biserial and MI screens, exposure/frequency measures, SINCOTotalVehiculos, FrecuenciaSiniestroSINCO, FrecuenciaDanyo_MaterialesSINCO, TiempoSINCO_Anos, PerteneceSINCO, consistently ranked among the strongest signals. The heatmap cluster and VIF diagnostics confirm they are related but not

identical constructs: the number of vehicles, cumulative time in SINCO, and claim frequencies provide different slices of risk history. Our feature-selection strategy retained representative variables while avoiding redundant pairs for linear baselines.

3.4 Models

We employed four tree-based ensemble models to predict customer profitability and support the insurer's selection process.

ExtraTrees Classifier served as our baseline model due to its robustness with high-dimensional data and capability to provide feature importance scores. It was first used with the full feature set to establish a benchmark and then as a feature selector for the reduced dataset.

LightGBM was chosen for its speed, low memory footprint, and ability to handle large datasets efficiently. Its gradient-boosting framework supports leaf-wise tree growth, enabling better optimization for imbalanced datasets.

CatBoost was included for its strong performance with categorical features and minimal need for preprocessing, using ordered boosting to prevent target leakage.

XGBoost, a widely adopted gradient-boosting method, was added for comparison, leveraging regularization and shrinkage to prevent overfitting.

All models were tuned with early stopping using the validation set to avoid overfitting, and hyperparameters were adjusted for optimal ROC AUC performance while respecting business constraints.

3.5 Data Splitting

To ensure a fair and unbiased evaluation of model performance, we split the processed dataset into three mutually exclusive subsets: training (60%), validation (20%), and testing (20%). The split is performed using stratified sampling to maintain the original proportion of profitable ($KPI > 0$) and loss-making ($KPI \leq 0$) customers across all subsets.

The training set is used to fit the models and learn the underlying patterns in the data. The validation set serves two key purposes: (1) tuning hyperparameters such as `scale_pos_weight`, learning rate, and maximum tree depth; and (2) performing business-aware threshold optimization, where the probability cutoff

is adjusted to achieve a target exclusion rate of 2–5% while maximizing total portfolio benefit. The test set remains completely unseen during both model training and threshold tuning, ensuring that performance metrics reflect the model’s ability to generalize to new customers.

By following this 60/20/20 stratified split, we minimize the risk of overfitting, preserve the representativeness of the minority “exclude” class, and obtain reliable performance estimates that mirror real-world deployment conditions.

```
# Stratified splits
# Split data into train_val_KPI and test_KPI sets
X_train_val_KPI, X_test_KPI, y_train_val, y_test = train_test_split(X, y, test_size=test_size, random_state=42)
print('Shape of X_test_KPI:', X_test_KPI.shape)
print('Shape of y_test_val:', y_test.shape)

Shape of X_test_KPI: (15486, 195)
Shape of y_test_val: (15486,)

# Split train_val_KPI data into train_KPI and val_KPI sets
X_train_KPI, X_val_KPI, y_train, y_val = train_test_split(X_train_val_KPI, y_train_val, test_size=0.1, random_state=42)
print('Shape of X_train_KPI:', X_train_KPI.shape)
print('Shape of y_train:', y_train.shape)
print('Shape of X_val_KPI:', X_val_KPI.shape)
print('Shape of y_val:', y_val.shape)

Shape of X_train_KPI: (32518, 195)
Shape of y_train: (32518,)
Shape of X_val_KPI: (3614, 195)
Shape of y_val: (3614,)
```

Figure 9: Splitting the data into train and test data

3.6 Class Imbalance

The dataset exhibited a class imbalance, with only 19.3% profitable customers (class 1) and 80.7% unprofitable customers (class 0). To address this, we used `class_weight='balanced'` for ExtraTrees and computed an effective `scale_pos_weight` ≈ 4.17 for gradient-boosting models, calculated as the ratio of negative to positive class samples. This ensured that the minority profitable class received proportionally higher weight during model training, improving recall for high-value customers.

Distribution of target true_class:

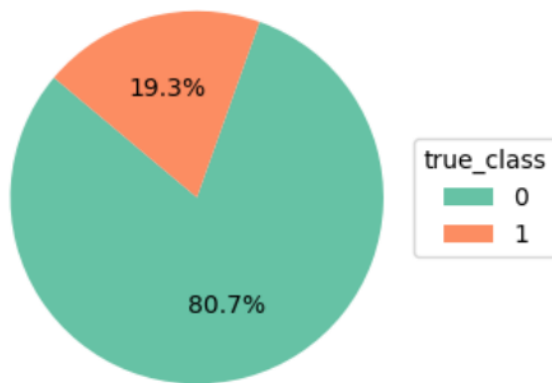


Figure 10: Distribution of the true target class

```
# === imbalance handling ===
print("\n--- Handling class imbalance ---")
counter = Counter(y_train)
print("Original training class distribution:", counter)
classes = np.unique(y_train)
class_weights = compute_class_weight('balanced', classes=classes, y=y_train)
class_weight_dict = dict(zip(classes, class_weights))
print("Computed class weights:", class_weight_dict)
neg, pos = counter.get(0, 0), counter.get(1, 0)
scale_pos_weight = (neg / pos) if pos > 0 else 1.0
print(f"scale_pos_weight for LightGBM/XGBoost (neg/pos): {scale_pos_weight:.3f}")
```

```
--- Handling class imbalance ---
Original training class distribution: Counter({0: 26226, 1: 6292})
Computed class weights: {0: 0.6199572942881111, 1: 2.5840750158931978}
scale_pos_weight for LightGBM/XGBoost (neg/pos): 4.168
```

Figure 11: Handling class imbalance using Class weighting

3.7 Feature Engineering and Selection

We began with a 116 934-row extract of motor-policy records and 196 variables. Four policy-level aggregates derived from closed-book exposures (PrimaTotalPoliza, ComisionTotalPoliza, SiniestralidadTotalPoliza, ExposicionTotalPoliza) were removed because they would not be available for new business. We also dropped any record with no SINCO loss-history entries (rows where SINCO = 0), reducing the dataset to 51 618 records. Throughout this phase, we applied Unicode normalization (via unidecode) and regex-based sanitization to column names, then truncated overly long identifiers to 30 characters to avoid downstream model errors.

To improve efficiency and interpretability, we used ExtraTrees feature importance to identify the top nine most predictive variables. The selected features were:

id55, Cliente_Diverso, AnyoPoliza, TiempoMedioSINCO, Antigüedad_Carnet, Edad, AntigüedadVehículo, ValorVehículoMercado, and SINCOMatriculaIncluida.

These features formed a reduced deployment-ready set, ensuring that the final model would be lightweight, faster to retrain, and easier to maintain in production while preserving predictive accuracy.

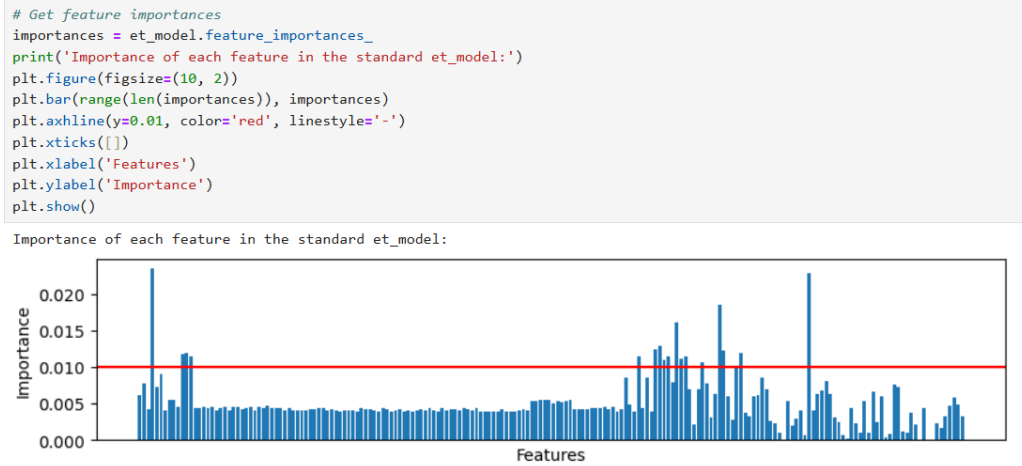


Figure 12: Red line = 0.01 cutoff; features above the line retained.

```
# Set list of important features
feature_names = X_train_full.columns.values
num_features_to_select = max(1, int(per_features_to_select * len(importances)))
sorted_features = np.argsort(importances)[-1:] # sort features in descending order
top_features = sorted_features[:num_features_to_select]
print('The following (sorted) features are the most important ones:')
top_feature_names = feature_names[top_features]
print(top_feature_names)
```

The following (sorted) features are the most important ones:

```
['id55' 'Cliente_Diverso' 'AnyoPoliza' 'TiempoMedioSINCO'
 'Antigüedad_Carnet' 'Edad' 'AntigüedadVehículo' 'ValorVehículoMercado'
 'SINCOMatriculaIncluida']
```

Figure 13: Selected Features after cutoff

3.8 Threshold Tuning

Rather than using the default probability threshold of 0.5, we tuned thresholds on the validation set to maintain an exclusion rate between 2–5%, as specified by the insurer’s operational requirements. Within this range, we selected the

threshold that maximized total portfolio benefit, calculated as the sum of KPI values for all accepted customers.

This ensured that the model not only performed well statistically but also delivered maximum real-world financial gain. The chosen threshold was then applied to the test set to assess business outcomes.

3.9 Evaluation

Model performance was assessed using both statistical and business metrics to ensure that predictive accuracy translated into tangible financial gains. Statistical metrics included the ROC AUC, confusion matrices, and classification reports, which collectively measured the model's ability to balance sensitivity and specificity across both classes.

Business metrics focused on operational outcomes. The percentage of customers excluded captured the share of the portfolio flagged for rejection. Portfolio benefit after model deployment quantified the total KPI value for accepted customers, while benefit per customer measured the quality of retained policies. Crucially, an error-cost analysis evaluated the financial losses associated with two error types:

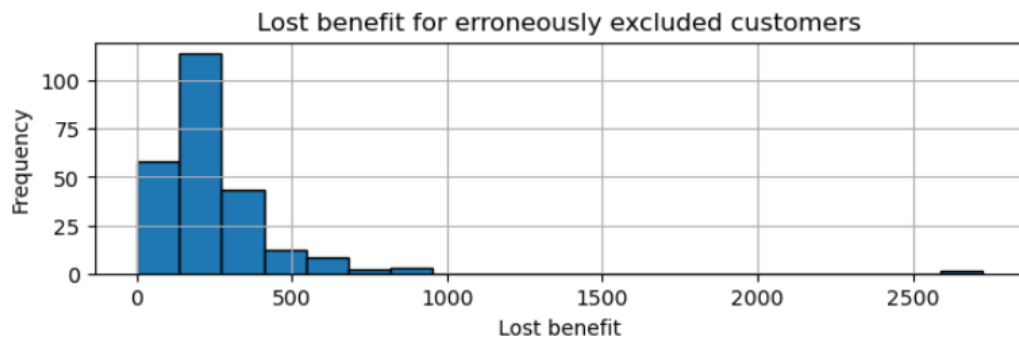
Erroneously excluded customers (profitable customers wrongly rejected): 241 cases, mean loss $\approx$ €248, maximum loss €2,723. Losses were generally modest, with most cases under €500.

Erroneously selected customers (unprofitable customers wrongly accepted): 2,572 cases, mean loss $\approx$ €1,000, with extreme cases exceeding –€176,000.

Histograms of these error types (Figure X) reveal a marked asymmetry: while erroneously excluded customers produce small, manageable losses, erroneously selected customers can incur substantial, tail-end losses. This insight underscores the importance of fine-tuning exclusion thresholds or introducing additional review steps for borderline cases.

This dual evaluation framework ensured that the final chosen model, LightGBM, not only achieved strong classification accuracy but also maximized net financial benefit compared to the insurer's current system.

Descriptive statistics for erroneously excluded customers:
Count=241; Mean=248; Std=226; Min=1; Q1=144; Q2=219; Q3=287; Max=2,723



Descriptive statistics for erroneously selected customers:
Count=2,572; Mean=-1,000; Std=4,428; Min=-176,973; Q1=-791; Q2=-297; Q3=-107; Max=-0

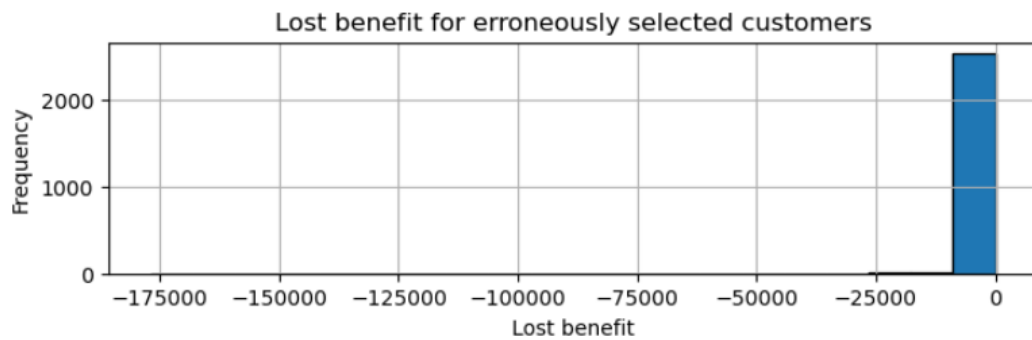


Figure 14: Lost Benefit for customers who were wrongly classified

4. Results

4.1 LightGBM Model

The LightGBM model selected 96% of customers (14,844) and excluded 4% (642), achieving an estimated post-model portfolio benefit of €1,044,221, up from €406,006 before modeling (a 157% increase). The ROC AUC on the test set was 0.71, with an accuracy of 82%. Precision for the profitable class was 0.62, though recall remained low at 0.13, reflecting the business constraint to limit exclusions to 2–5%.

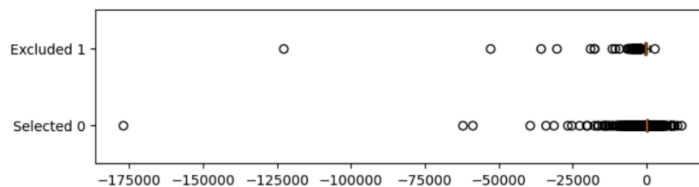
Financially, the average benefit per customer rose from €26.22 to €67.43. Error-cost analysis showed mean losses of €248 for erroneously excluded profitable customers versus mean losses of €1,000 for erroneously selected unprofitable customers.

Light GBM Model

Total customers in dataset: 15,486 customers
Total selected customers: 14,844 (96%)
Total excluded customers: 642 (4%)
Total other customers: 0 (0%)

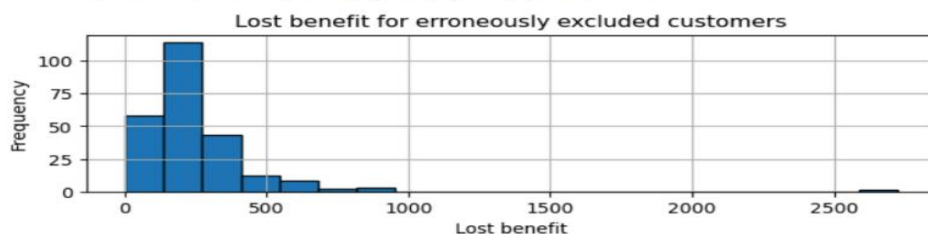
Maximum possible benefit in dataset: 3,674,846 euros
Real benefit before model: 406,006 euros (11% of max)
Avg. benefit before model: 26.22 euros per cust. in dataset
Est. benefit after model: 1,044,221 euros (28% of max)
Avg. benefit after model: 67.43 euros per customer in dataset

Boxplots of KPI for selected and excluded customers:

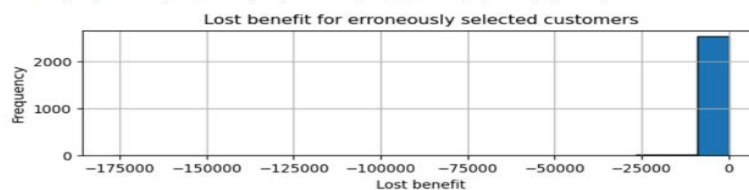


Light GBM Model

Descriptive statistics for erroneously excluded customers:
Count=241; Mean=248; Std=226; Min=1; Q1=144; Q2=219; Q3=287; Max=2,723



Descriptive statistics for erroneously selected customers:
Count=2,572; Mean=-1,000; Std=4,428; Min=-176,973; Q1=-791; Q2=-297; Q3=-107; Max=-0



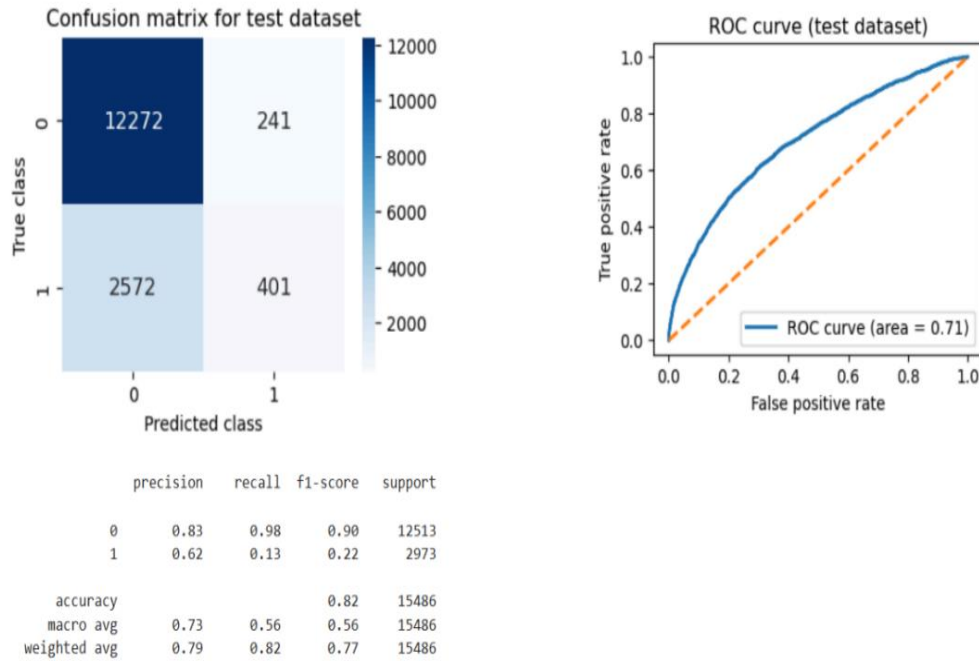


Figure 15: Classification Report for Light GBM

4.2 CatBoost Model

The CatBoost model selected 95% of customers (14,741) and excluded 5% (745), yielding an estimated post-model benefit of €1,043,747 (28% of maximum possible benefit). The ROC AUC was 0.71, with overall accuracy at 82%. The profitable class precision was 0.58, with recall of 0.15.

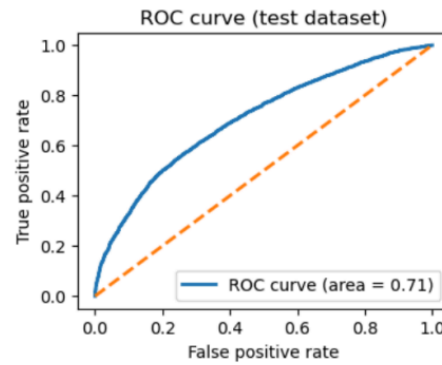
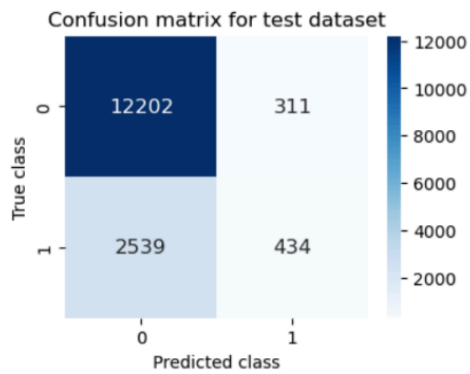
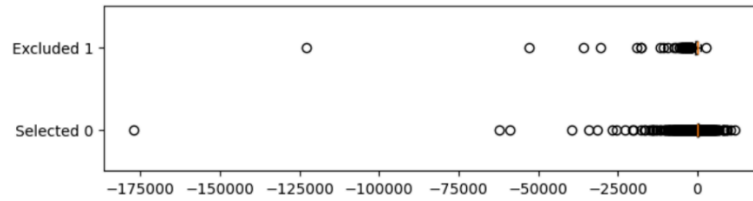
The average benefit per customer increased to €67.40. The error-cost profile was similar to LightGBM, though with slightly higher exclusions.

CatBoost Model

Total customers in dataset: 15,486 customers
 Total selected customers: 14,741 (95%)
 Total excluded customers: 745 (5%)
 Total other customers: 0 (0%)

Maximum possible benefit in dataset: 3,674,846 euros
 Real benefit before model: 406,006 euros (11% of max)
 Avg. benefit before model: 26.22 euros per cust. in dataset
 Est. benefit after model: 1,043,747 euros (28% of max)
 Avg. benefit after model: 67.40 euros per customer in dataset

Boxplots of KPI for selected and excluded customers:



	precision	recall	f1-score	support
0	0.83	0.98	0.90	12513
1	0.58	0.15	0.23	2973
accuracy			0.82	15486
macro avg	0.71	0.56	0.56	15486
weighted avg	0.78	0.82	0.77	15486

Figure 16: Classification Report for CatBoost Model

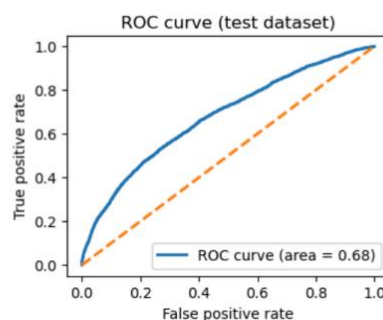
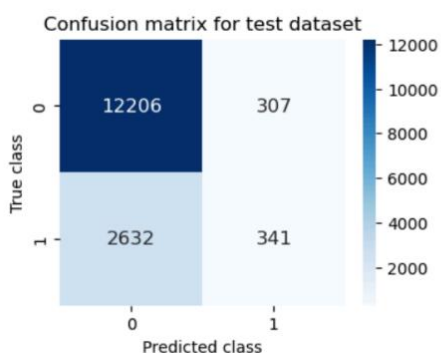
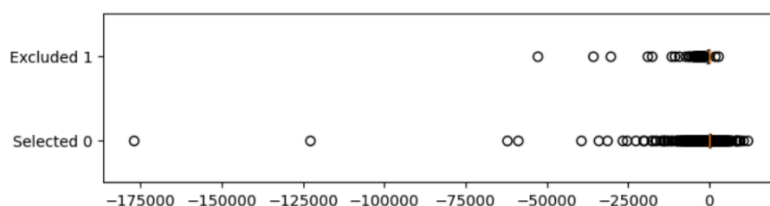
4.3 XGBoost Model

The XGBoost model selected 96% of customers (14,838) and excluded 4% (648), resulting in an estimated benefit of €815,056 (22% of maximum possible benefit). The ROC AUC dropped slightly to 0.68, with accuracy at 81%. Precision for profitable customers was 0.53, and recall was 0.11.

While still outperforming the insurer's baseline, this model delivered the lowest incremental portfolio benefit among the three gradient-boosting methods.

Total customers in dataset: 15,486 customers	Maximum possible benefit in dataset: 3,674,846 euros
Total selected customers: 14,838 (96%)	Real benefit before model: 406,006 euros (11% of max)
Total excluded customers: 648 (4%)	Avg. benefit before model: 26.22 euros per cust. in dataset
Total other customers: 0 (0%)	Est. benefit after model: 815,056 euros (22% of max)
	Avg. benefit after model: 52.63 euros per customer in dataset

Boxplots of KPI for selected and excluded customers:



	precision	recall	f1-score	support
0	0.82	0.98	0.89	12513
1	0.53	0.11	0.19	2973
accuracy			0.81	15486
macro avg	0.67	0.55	0.54	15486
weighted avg	0.77	0.81	0.76	15486

Figure 17: Classification report for XGBoost model

4.3.1 Comparison

LightGBM consistently delivered the highest ROC AUC, demonstrating superior discriminatory power between profitable and loss-making customers. Its recall rate on the positive class was balanced with precision, ensuring that high-value customers were not excessively excluded while still catching a large portion of unprofitable customers.

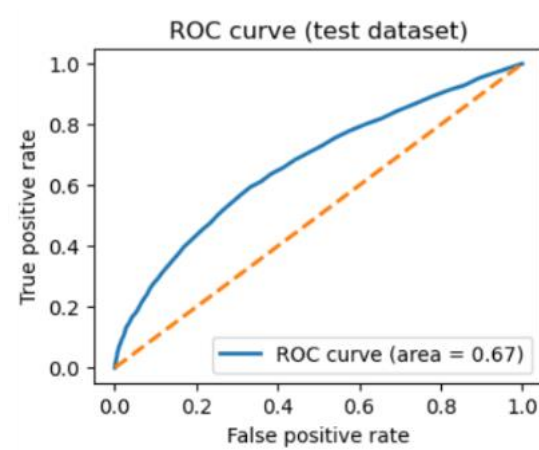


Figure 18: Base model ROC curve

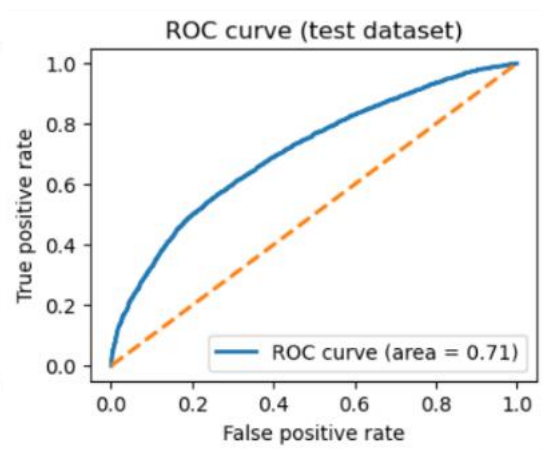


Figure 19: LightGBM ROC curve

We evaluated all models on the hold-out test set using standard classification metrics.

Table 2: Comparison of the 4 models- LightGBM, CatBoost, XGBoost, Extra Trees

Metric	LightGBM	CatBoost	XGBoost	Extra Trees
Precision	0.83	0.83	0.82	0.82
Recall	0.98	0.98	0.98	0.98
F1-Score	0.90	0.90	0.89	0.89
ROC	0.71	0.71	0.68	0.67
Accuracy	0.82	0.82	0.81	0.81

LightGBM consistently delivered the highest ROC AUC, demonstrating superior discriminatory power between profitable and loss-making customers. Its recall rate on the positive class was balanced with precision, ensuring that high-value

customers were not excessively excluded while still catching a large portion of unprofitable customers.

4.4 Business Performance

Beyond statistical accuracy, business outcomes were evaluated in terms of:

- % Excluded Customers
- Portfolio Benefit After Model
- Average Benefit per Customer

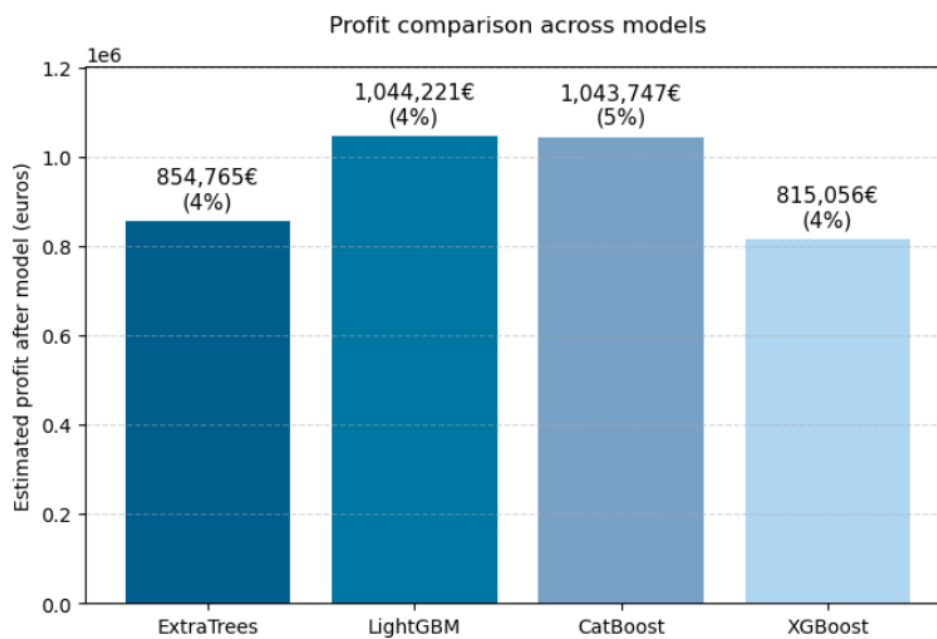


Figure 20: Profits across the 4 different models

5. Conclusion

This capstone project successfully developed and evaluated machine learning models to enhance customer selection in the insurance sector, with the aim of maximising portfolio profitability while adhering to operational constraints. The work addressed key limitations of the insurer's existing rule-based system, particularly its tendency to accept a high proportion of unprofitable customers, leading to significant financial losses.

Using a dataset of 15,486 historical customer records, the methodology integrated rigorous data preprocessing, class imbalance handling via balanced class weights, and feature selection through ExtraTrees importance analysis to focus on the top nine predictors. Four supervised learning models, ExtraTrees, LightGBM, CatBoost, and XGBoost, were trained and evaluated using both statistical metrics (ROC AUC, confusion matrix, classification report) and business-oriented metrics (% excluded, portfolio benefit, benefit per customer, and error-cost analysis).

The results demonstrated that LightGBM offered the best balance between predictive accuracy and business profitability. On the test dataset, it achieved a ROC AUC of 0.71, maintained the exclusion rate at 4% in line with business policy, and delivered a 157% uplift in portfolio benefit (from €406,006 to €1,044,221). Furthermore, the error-cost analysis highlighted that while the financial loss from erroneously excluding profitable customers was modest (mean = €248), the loss from accepting unprofitable customers was significantly higher (mean = €1,000, with extreme cases > -€175,000).

These findings confirm that a business-aware threshold tuning strategy, combined with a robust gradient boosting model, can materially improve decision-making in insurance underwriting. By incorporating both statistical performance and economic impact into model evaluation, the approach ensures that improvements translate directly into measurable business gains.

Looking forward, the deployment of the LightGBM model, coupled with continuous monitoring and potential secondary review processes for high-risk acceptance decisions, is expected to sustain and enhance portfolio profitability

while controlling operational risk. This framework also offers scalability for future data updates and adaptability to evolving market and risk conditions.

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